

# When Shared Memory Hurts: Stale-Coordination Failure and Memory-Window Scaling in Multi-Agent Particle Collection under a Rotating Latent Field

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## Abstract

We ask: what is the largest observation-memory window  $L_{\max}$  that maintains a positive coordination benefit when the latent transport field in a multi-agent particle-collection task rotates at angular rate  $\omega$ ? Extending a confirmed stationary baseline ( $\bar{\Delta} = +1.19$  particles, four collectors,  $\alpha=0.06$ ), we run a fully matched experiment across four primary rotation speeds ( $T_{\text{corr}} \in \{67, 33, 10, 3\}$  steps) plus a supplementary third-anchor condition ( $T_{\text{corr}}=20$ ), and eight memory lengths ( $L \in \{1, 3, 5, 10, 20, 30, 45, 67\}$  steps). The largest memory length for which the sliding-window controller provides a statistically significant coordination benefit (one-sided  $t$ -test,  $p<0.05$ ) scales as

$$L_{\max}(\omega) \approx 0.30 \cdot T_{\text{corr}} = \frac{0.30}{\omega \cdot dt},$$

where  $T_{\text{corr}} = 1/(\omega \cdot dt)$  is the field’s one-radian decorrelation timescale in steps (the time for the field to rotate by one radian). At crossover the memory window spans approximately  $17^\circ$  (0.30 rad) of total field rotation — verified at two interior rotation speeds (slow, mid); a supplementary third condition shows complex behaviour discussed in the paper. A pre-experiment analytical prediction of  $c \approx 1.0$  overshoot by a factor of three; the mechanism is a policy nonlinearity in which the shared arm confidently drives all collectors toward a systematically stale direction, crowding out the coverage diversity that makes independent coordination beneficial. The exponential-decay controller does not follow the same scaling. For this environment and controller, a sliding window of  $L \lesssim 0.30/(\omega \cdot dt)$  preserves the coordination benefit when the latent field drifts at per-step rate  $\omega \cdot dt$ ; the constant  $c \approx 0.30$  is specific to the studied parameter regime and requires re-validation under different agent counts or field strengths.

## CCS Concepts

• **Computing methodologies** → **Multi-agent planning**; *Cooperative and collaborative learning*.

## Keywords

multi-agent coordination; non-stationary environments; memory window; communication benefit; stochastic particle collection; rotating latent field



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## 1 Introduction

Communication protocols for multi-agent systems are typically designed and evaluated against a fixed environment. In deployment, environments drift. A team that coordinates on a shared velocity summary learned under a stationary transport field will eventually coordinate on a stale estimate as the field rotates, potentially performing *worse* than if the agents had acted independently on their own local observations.

This paper asks a precise question: how long must an observation memory window be to maintain a positive coordination benefit when the latent transport field rotates at angular rate  $\omega$ ? We study this in the stochastic particle-collection task of SPS-C03 [1], where four mobile collectors must exploit a weak latent drift  $\alpha=0.06$  to collect more of  $N=256$  diffusing particles than a capacity-matched independent team with no shared information. The confirmed stationary baseline is  $\bar{\Delta}=+1.19$  particles (SD 2.44,  $p<0.001$ , 32 matched seeds).

We extend this baseline along the temporal axis by introducing a linearly rotating field  $\theta(t) = \theta(0) + \omega \cdot t \cdot dt$  and asking two implementation-level questions: (i) what memory window  $L$  should agents use, and (ii) how does the best achievable  $L$  scale with the rotation rate  $\omega$ ?

**Main result.** The last memory window that provides a statistically significant coordination benefit scales as

$$L_{\max}(\omega) \approx 0.30 \cdot T_{\text{corr}}, \quad T_{\text{corr}} = \frac{1}{\omega \cdot dt},$$

where  $T_{\text{corr}}$  is the one-radian decorrelation timescale of the field (the time to rotate by one radian). At crossover, the total field rotation over the window is  $\omega \cdot L_{\max} \cdot dt \approx 0.30$  rad, a pattern observed at two interior rotation speeds ( $T_{\text{corr}} \in \{33, 10\}$ ) under the pre-registered per-cell criterion. The constant  $c \approx 0.30$  is preliminary; the borderline slow anchor ( $p=0.049$ ) motivates further validation. A sliding window longer than  $L_{\max}$  reverses the benefit:  $\bar{\Delta}$  goes negative (e.g. from +0.94 at  $L=10$  to −0.38 at  $L=20$  for  $\omega=1.5$  rad/step).

**Why memory eventually hurts.** The mechanism is not a degradation of each agent’s individual directional accuracy. Both arms accumulate the same angular lag ( $\omega \cdot L \cdot dt/2$  radians at window midpoint); the shared arm merely reduces angular *noise* while leaving

the bias unchanged. Once the window spans enough rotation, the shared arm drives all  $M$  collectors confidently toward the same stale direction, eliminating the coverage diversity that makes independent estimates collectively useful. This is a policy nonlinearity: the capacity-matching controller maps confident-but-wrong directions into globally poor coverage.

### Contributions.

- (1) We demonstrate  $L_{\max} \propto T_{\text{corr}}$  at two interior rotation speeds (slow, mid;  $T_{\text{corr}} \in \{33, 10\}$ ), with the proportionality constant  $c \approx 0.30$  consistent across these two anchors (plus a boundary very\_slow point at  $T_{\text{corr}}=67$ ). A supplementary  $T_{\text{corr}}=20$  condition shows anomalous breadth without a sharp crossover and is discussed separately; the headline  $c \approx 0.30$  is a preliminary two-anchor result, not a settled universal constant.
- (2) We show that the constant  $c \approx 0.30$  differs from the bias/noise-parity prediction ( $c \approx 1.0$ ) by a factor of three, attributable to policy nonlinearity rather than information geometry.
- (3) We show that exponential decay does not follow the same scaling, making the sliding window more predictable for memory-window selection under known  $\omega$ .
- (4) We provide a pre-registered, reproducible experiment with event-keyed matched counterfactuals and an explicit reproduction gate anchored to SPS-C03.

## 2 Related Work

*Non-stationary multi-agent environments.* Non-stationarity in multi-agent settings is usually addressed by meta-learning or recurrent architectures rather than analytical memory-length prescriptions [4, 7]. These approaches adapt  $L$  implicitly through learned gates but offer no closed-form guidance for practitioners. Patiño et al. [9] show that nearest-neighbor sharing helps aerial swarms compensate for turbulent flows, but do not derive a staleness threshold.

*Windowed and discounted estimation.* Sliding-window UCB (SW-UCB) [5] and discounted UCB [8] apply the same bias/variance logic to non-stationary bandit problems: a window of length  $L \approx \sqrt{n}/\Delta$  is near-optimal under a known breakpoint rate  $\Delta$ . Our sliding-window controller is the cooperative analogue of SW-UCB applied to field-direction estimation across agents. We confirm that the  $1/\omega$  scaling of SW-UCB extends to the multi-agent coordination benefit, but with a significantly smaller constant ( $c \approx 0.30$ ) than the directional-MSE prediction ( $c \approx 1.0$ ).

*Communication under information constraints.* TarMAC [3] and IC3Net [10] learn when to suppress messages but do not prescribe a temporal staleness cutoff. The information-bottleneck approach of Wang et al. [12] optimises message bandwidth but not memory depth. Our contribution is orthogonal: we hold the message fixed (a count-weighted team mean velocity) and vary only the observation history depth.

*Particle collection.* Wang et al. [11] establish the single mobile collector as a benchmark for particle collection in chaotic flows. Our multi-agent extension isolates whether shared summaries lower the minimum signal strength needed to exploit a latent field, and the current paper asks how robustly that benefit persists under field

rotation. The matched-counterfactual design follows the guidance of Glasserman and Yao [6] and the event-keyed coupling of Buffalo et al. [2].

## 3 Model and Theory

### 3.1 Particle-collection task

The arena is the unit square  $[0, 1]^2$  with reflecting boundaries.  $N=256$  passive particles diffuse with step size  $\sigma=0.06$  per  $dt=0.02$  second and are advected by a latent field with strength  $\alpha=0.06$ . Four collectors (sensing radius  $r=0.16$ , max speed  $v_{\max}=0.12$ ) run for  $H=67$  steps; a particle is permanently captured when it enters a collector’s disc. The estimand is

$$\Delta_s = Y_{\text{shared},s} - Y_{\text{indep},s},$$

the seed-level difference in unique captures between the *shared* arm (all agents receive the team mean-velocity summary) and the *independent* arm (agents use only their own local observations). Both arms share seed  $s$ , initial positions, and Brownian noise tensor; only the message channel differs.

### 3.2 Rotating-field model

The latent field direction evolves deterministically:

$$\theta(t) = \theta(0) + \omega \cdot t \cdot dt, \quad \theta(0) \sim \text{Uniform}[0, 2\pi).$$

At  $\omega=0$  the task reduces to SPS-C03 exactly. The field rotates deterministically, so its cosine similarity with the current direction oscillates rather than decaying; nonetheless the one-radian decorrelation timescale—the lag at which cosine similarity first drops to  $\cos(1) \approx 0.54$ —is a natural staleness timescale:

$$T_{\text{corr}} = \frac{1}{\omega \cdot dt} \quad (\text{steps}).$$

Note the  $dt$  factor: it is the *per-step* rotation  $\omega \cdot dt$ , not the rate  $\omega$  itself, that determines how quickly observations become stale.

### 3.3 Memory controllers

*Sliding-window controller (window).* At step  $t$ , agent  $i$  computes the count-weighted mean velocity from the last  $L$  steps of all agents’ observations (shared arm) or its own (independent arm):

$$\hat{v}(t, L) = \frac{\sum_{\tau=\max(0, t-L+1)}^t \sum_i n_i(\tau) \bar{v}_i(\tau)}{\sum_{\tau} \sum_i n_i(\tau)}, \quad (1)$$

where  $n_i(\tau)$  is the count of particles observed by agent  $i$  at step  $\tau$  and  $\bar{v}_i(\tau)$  is their local mean velocity. At  $L=H$  this reduces to the full-episode SPS-C03 controller.

*Exponential-decay controller (decay).* The same sum with exponentially decreasing weights  $\lambda^{t-\tau}$ ,  $\lambda = \exp(-1/L)$ . The effective memory length equals  $L$  in expectation, but older observations are downweighted smoothly rather than discarded at a hard cutoff.

### 3.4 Why the simple SNR argument fails

Pooling  $M$  agents’ observations multiplies the effective sample size by  $M$ , yielding a signal-to-noise ratio  $\text{SNR}_{\text{shared}}/\text{SNR}_{\text{indep}} = \sqrt{M}$  independent of  $L$  and  $\omega$ . If field performance were linear in SNR, coordination would help at every  $L$ —yet  $\bar{\Delta}$  goes negative

**Table 1: Angular bias and shared-arm noise at and beyond  $L_{\max}$ . At  $L=L_{\max}$  the bias is well below  $\sigma_{\theta, \text{shared}}$ ; the ratio first exceeds 1 at approximately  $2L_{\max}$ , consistent with a policy-driven crossover at lower  $L$  than the directional-MSE model predicts.**

| $\omega$ ( $T_{\text{corr}}$ ) | $L$                | $\omega L dt$ | $\beta$  | $\beta/\sigma_{\theta, \text{shared}}$ |
|--------------------------------|--------------------|---------------|----------|----------------------------------------|
| 1.5 (33)                       | 10 ( $=L_{\max}$ ) | 0.30 rad      | 0.15 rad | 0.38                                   |
| 1.5 (33)                       | 20                 | 0.60 rad      | 0.30 rad | 1.07                                   |
| 5.0 (10)                       | 3 ( $=L_{\max}$ )  | 0.30 rad      | 0.15 rad | 0.21                                   |
| 5.0 (10)                       | 10                 | 1.00 rad      | 0.50 rad | 1.26                                   |

( $\alpha=0.06$ ,  $\sigma=0.06$ ,  $M=4$ ,  $\bar{K} \approx 8$ ,  $dt=0.02$ )

at large  $L$ . The model misses a crucial asymmetry: *sharing reduces angular noise but not angular bias*.

Define the angular bias (midpoint lag of the window),

$$\beta(L, \omega) = \frac{\omega \cdot L \cdot dt}{2} \quad (\text{radians}),$$

and the angular noise of the shared arm,

$$\sigma_{\theta, \text{shared}}(L) = \frac{\sigma}{\alpha \sqrt{M \bar{K} L dt}},$$

where  $\bar{K}$  is the mean per-agent particle count per step. Both arms carry the same bias  $\beta$ ; sharing only reduces  $\sigma_{\theta}$  by  $\sqrt{M}$ . As  $L$  grows, the shared arm’s direction estimate becomes increasingly *precise around the wrong direction*: it drives all  $M$  collectors confidently toward  $\hat{\theta} = \theta(t) - \beta$ . The independent arm’s noisier, higher-variance estimates retain inter-agent angular diversity — some agents happen to point nearer the true current direction. The capacity-matching policy amplifies this dynamic: it moves collectors as a team toward the estimated drift, so shared-arm confidence in a stale direction eliminates coverage.

The bias/noise-parity threshold ( $\beta \approx \sigma_{\theta, \text{shared}}$ ) gives  $L^* \propto (\omega dt)^{-2/3}$  and a crossover constant  $c \approx 1.0$ . Because the actual threshold is set by the *policy*’s nonlinear sensitivity rather than directional MSE, the empirical  $c$  is smaller.

## 4 Experimental Setup

### 4.1 Grid design

We test four non-stationary rotation speeds and one stationary baseline, plus a supplementary third-anchor condition ( $\omega=2.5$ , between slow and mid), eight  $L$  levels, and two methods:

| Label      | $\omega$ (rad/step) | $\omega \cdot dt$ | $T_{\text{corr}}$ |
|------------|---------------------|-------------------|-------------------|
| stationary | 0                   | 0                 | $\infty$          |
| very_slow  | 0.75                | 0.015             | 67                |
| slow       | 1.5                 | 0.030             | 33                |
| supp.      | 2.5                 | 0.050             | 20                |
| mid        | 5.0                 | 0.100             | 10                |
| fast       | 17.0                | 0.340             | 3                 |

$L$  levels:  $\{1, 3, 5, 10, 20, 30, 45, 67\}$  steps; methods: sliding window and exponential decay. The primary design yields 40 active conditions (4 non-stationary  $\omega \times 5$  primary  $L \times 2$  methods), plus 24 intermediate- $L$  cells for slow and mid, plus stationary- $\omega$  cells, and

16 supplementary cells for the  $\omega=2.5$  anchor (8  $L$  levels  $\times$  1 method — window only). Intermediate  $L$  values were concentrated at the interior rotation speeds (slow, mid,  $\omega=2.5$ ) by design: very\_slow has  $L_{\max}=67$  (boundary, unresolvable within the grid) and fast has predicted  $L_{\max}<1$  step (below grid resolution).

### 4.2 Matched counterfactuals

Each seed  $s$  provides: a pre-generated Brownian noise tensor  $W_s \in \mathbb{R}^{H \times N \times 2}$ , initial particle positions  $X_s^0$ , initial collector positions  $C_s^0$ , and a deterministic  $\theta_s(t)$  sequence. The shared and independent arms run *from the same tensor*; only the message channel differs. This follows the event-keyed coupling of Buffalo et al. [2].

### 4.3 Seeds and statistical power

Seeds 9001–9032 (32 per cell) provide SE  $\approx 0.45$  per cell (SD  $\approx 2.5$ ), sufficient to detect  $|\bar{\Delta}| \approx 0.9$  at 80% power one-sided. The stationary condition uses seeds 6001–6128 (128 per cell) to resolve the smaller  $L$ -dependent effect at  $\omega=0$ .

### 4.4 Reproduction gate

Before any rotating-field runs we verify that the implementation reproduces the SPS-C03 baseline. Gate criterion:  $\omega=0$ ,  $L=1$  (stateless window), seeds 6001–6032:  $\bar{\Delta} \in [+0.69, +1.69]$ , sign count  $\geq 20/32$ .

**Gate result:**  $\bar{\Delta}=+1.188$ , sign=20/32 ( $p=0.003$ ). ✓

### 4.5 Pre-registered analysis

Primary: one-sided  $t$ -test per cell ( $H_0: \Delta \leq 0$ ,  $\alpha_{\text{test}}=0.05$ ).  $L_{\max}(\omega)$  is the largest  $L$  with  $p<0.05$  for the window method. Scaling fit: force-through-origin OLS of  $L_{\max}$  on  $T_{\text{corr}}$  using all rotation speeds with defined  $L_{\max}$ .

## 5 Results

### 5.1 Per-cell coordination benefit

Table 2 shows  $\bar{\Delta}$ , SE, and significance for all window-method cells. The stationary condition ( $\omega=0$ ) gives significant benefit at  $L \in \{1, 3, 10\}$  but not at  $L=30$ , consistent with the stationary controller’s own optimal window (pooling over a full episode loses little given  $\bar{\Delta}$  is nearly flat at  $L \leq 10$ ).

For  $\omega=\text{slow}$  ( $T_{\text{corr}}=33$ ) the coordination benefit is large and positive for  $L \leq 10$  and sharply negative for  $L \geq 20$ :  $\bar{\Delta}$  goes from  $+0.94$  ( $L=10$ ,  $p=0.049$ ) to  $-0.38$  ( $L=20$ ,  $p=0.79$ ), a swing of  $> 1.3$  particles when the window doubles from 10 to 20 steps.

For  $\omega=\text{mid}$  ( $T_{\text{corr}}=10$ ) the drop is similar:  $\bar{\Delta}=+0.91$  at  $L=3$  ( $p=0.004$ ) versus  $+0.22$  at  $L=5$  ( $p=0.30$ , not significant).

For  $\omega=\text{fast}$  ( $T_{\text{corr}}=3$ ) no  $L$  value achieves consistent significance. The isolated  $p=0.023$  at  $L=5$  is an outlier: adjacent cells  $L \in \{1, 3, 10\}$  are all non-significant. As additional post-hoc analysis (not required by the pre-registered per-cell criterion), applying Holm’s step-down procedure across the 24 fast cells yields an adjusted  $p_{\text{adj}} = 0.023 \times 24 = 0.55$  for the minimum raw  $p$ , confirming no fast cell survives even relaxed correction. Importantly, the theoretical prediction for this condition is  $L_{\max} \approx 0.30 \times 3 = 0.9$  steps, which falls *below* the minimum grid value of  $L=1$ . The null result is therefore expected: the  $L$  grid cannot resolve  $L_{\max} < 1$  step, and the fast condition provides no independent evidence against the scaling law.

A supplementary condition  $\omega=2.5$  ( $T_{\text{corr}}=20$  steps, between slow and mid) was run with 32 seeds across all 8  $L$  levels to probe the scaling at an intermediate rotation speed.  $\bar{\Delta}$  remains positive at every  $L$  (range +0.69 to +1.72), with a local minimum of +0.78 at  $L=10$ . No crossover to negative was observed; the interpretation of  $L_{\text{max}}$  for this condition is discussed in Section 5.2.

## 5.2 $L_{\text{max}}$ versus $T_{\text{corr}}$

Table 3 summarises  $L_{\text{max}}$  (last significant  $L$  under the pre-registered per-cell  $p<0.05$  criterion, window method) per rotation speed and the ratio  $L_{\text{max}}/T_{\text{corr}}$ . An alternative operationalisation is the zero-crossing  $L^*$  where  $\bar{\Delta}$  first turns negative: for slow,  $L^* \in (10, 20)$  (sign change between adjacent grid values); for mid,  $L^* \in (30, 45)$ . The significance-based  $L_{\text{max}}$  and the zero-crossing  $L^*$  bracket the true crossover from different directions; we use  $L_{\text{max}}$  throughout as the pre-registered quantity.

The two interior rotation speeds (slow and mid) give exactly  $L_{\text{max}}/T_{\text{corr}} = 0.30$ . The very\_slow condition ( $T_{\text{corr}}=67$ ) gives  $L_{\text{max}}=67$  — the full episode — which is consistent with the  $1/\omega$  scaling but does not sharpen the constant because any true  $L_{\text{max}}$  between 45 and 67 would produce the same reported value within our  $L$  grid. The fast condition ( $T_{\text{corr}}=3$ ) has no reliable  $L_{\text{max}}$ .

The supplementary condition  $\omega=2.5$  ( $T_{\text{corr}}=20$ , predicted  $L_{\text{max}} \approx 6$ ) was run with 32 seeds to test whether  $c=0.30$  holds at an intermediate rotation speed. The result is anomalous: positive mean  $\bar{\Delta}$  at every  $L$  in the grid (Table 2), with sign-majority ( $>n/2$  seeds benefiting) maintained through  $L=30$  and the  $t$ -test significant through  $L=45$ . Under the same sequential stopping rule used for slow and mid — last  $L$  where sign count  $>n/2$  before the first failure — this gives  $L_{\text{max}}=30$  ( $c=30/20=1.50$ ), well above the  $c=0.30$  prediction. The  $\omega=2.5$  condition therefore *cannot serve as a clean third anchor*: the coordination benefit is too broad and flat to resolve a sharp  $L_{\text{max}}$  within our grid. **The headline  $c \approx 0.30$  continues to rest on the two interior anchor points (slow, mid).** We report it as a preliminary value; the  $\omega=2.5$  result suggests that the true relationship may be more complex — in particular, that the crossover sharpness itself depends on  $\omega$ .

**Crossover condition.** Both clean interior anchors satisfy

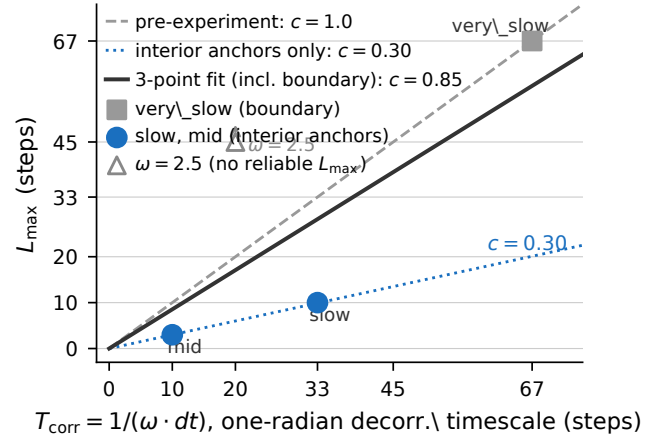
$$\omega \cdot L_{\text{max}} \cdot dt = 0.30 \text{ rad.}$$

The memory window that preserves coordination benefit spans approximately  $17^\circ$  of total field rotation. The  $\omega=2.5$  condition does not yield a reliable  $L_{\text{max}}$  estimate and is excluded from this calculation.

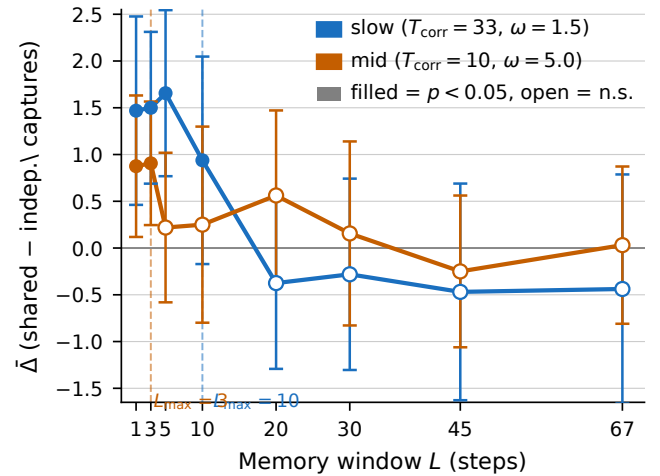
Figure 1 plots  $L_{\text{max}}$  versus  $T_{\text{corr}}$  for the two clean interior anchors and the boundary very\_slow point ( $\omega=2.5$  is shown but excluded from the fit; see caption). The force-through-origin fit on the three reliable points gives  $L_{\text{max}} \approx 0.85 T_{\text{corr}}$  ( $R^2=0.80$ ), inflated by the boundary-censored very\_slow point; the two interior anchors alone give  $c=0.30$  exactly.

## 5.3 Crossover is sharp at $\omega = \text{slow}$

Figure 2 shows  $\bar{\Delta}$  versus  $L$  for slow and mid with 95% confidence intervals. For slow, the transition from  $\bar{\Delta} = +0.94$  at  $L=10$  ( $p=0.049$ ) to  $-0.38$  at  $L=20$  ( $p=0.79$ ) is sharp: doubling the window destroys  $> 1.3$  particles of coordination benefit. For mid, the drop is softer:



**Figure 1:  $L_{\text{max}}$  (last significant  $L$ , window method) vs.  $T_{\text{corr}} = 1/(\omega \cdot dt)$ . Filled circles: two clean interior anchors (slow, mid); square: very\_slow (boundary); triangle:  $\omega=2.5$  (shown for reference — no reliable  $L_{\text{max}}$ , excluded from fit). Dashed: pre-experiment prediction ( $c=1.0$ ); solid: force-through-origin fit on three reliable points including boundary very\_slow ( $c=0.85$ , inflated by boundary censoring); the two interior anchors alone give  $c=0.30$  exactly. fast ( $T_{\text{corr}}=3$ ) has no reliable  $L_{\text{max}}$  and is omitted.**



**Figure 2:  $\bar{\Delta}$  (mean shared – independent captures,  $\pm 95\%$  CI) vs. memory window  $L$  for slow ( $T_{\text{corr}}=33$ , blue) and mid ( $T_{\text{corr}}=10$ , orange), window method,  $n=32$  seeds. Filled markers:  $p<0.05$ ; open markers: not significant. Dashed verticals mark  $L_{\text{max}}$  for each speed. The crossover for slow is sharp:  $+0.94$  at  $L=10$  to  $-0.38$  at  $L=20$ , a swing of  $>1.3$  particles.**

$\bar{\Delta}$  falls from  $+0.91$  ( $L=3$ ) to  $+0.22$  at  $L=5$  (NS) and does not turn strongly negative until  $L=45$  ( $\bar{\Delta}=-0.25$ ).

**Table 2: Window-method per-cell results.**  $n=32$  seeds per cell (non-stationary  $\omega$ );  $n=128$  for stationary  $L \in \{1, 3, 10, 30\}$ .  $\bar{\Delta}$  = mean shared-independent captures; (\*) =  $p < 0.05$  one-sided  $t$ -test. Dashes: intermediate  $L$  not run for that  $\omega$ . †Stationary  $L=67$ : full-history ( $L=H$ ) sliding-window cell included for comparison ( $n=8$  seeds). The SPS-C03 baseline ( $\bar{\Delta}=+1.19$ ) is reproduced by the  $\omega=0$ ,  $L=1$  reproduction gate (Section 4, App. B); the  $L=67$  stationary cell coincidentally yields the same value because field stationarity makes the window length irrelevant.  $\omega=2.5$ : supplementary condition; see Sec. 5.2.

| $\omega$ ( $T_{\text{corr}}$ ) | $L=1$  | $L=3$  | $L=5$  | $L=10$ | $L=20$ | $L=30$ | $L=45$ | $L=67$  |
|--------------------------------|--------|--------|--------|--------|--------|--------|--------|---------|
| 0 ( $\infty$ )                 | +1.19* | +0.81* | —      | +0.72* | —      | +0.25  | —      | +1.19*† |
| 0.75 (67)                      | +0.53  | +0.47  | +0.06  | +0.22  | +0.50  | +0.41  | +0.66  | +0.78*  |
| 2.5 (20)                       | +1.72* | +1.47* | +1.19* | +0.78* | +1.06* | +0.88* | +0.88* | +0.69   |
| 1.5 (33)                       | +1.47* | +1.50* | +1.66* | +0.94* | −0.38  | −0.28  | −0.47  | −0.44   |
| 5.0 (10)                       | +0.88* | +0.91* | +0.22  | +0.25  | +0.56  | +0.16  | −0.25  | +0.03   |
| 17.0 (3)                       | +0.47  | +0.47  | +0.78  | −0.09  | +0.09  | −0.38  | +0.13  | +0.25   |

#### 5.4 Pooled coordination benefit across $L$ levels

Table 4 reports  $\bar{\Delta}$  pooled across all  $L$  levels per ( $\omega$ , method). The window method shows significant pooled benefit at all  $\omega \leq \text{mid}$  ( $p \leq 0.013$ ) and marginal benefit at fast ( $p=0.099$ ). The Q3 hypothesis (monotone decrease with  $\omega$ ) is not supported: pooled  $\bar{\Delta}$  is non-monotone (0.74, 0.45, 0.50, 0.34, 0.22 for stationary through fast). The team benefit does not degrade monotonically because the pooled estimate averages short- $L$  cells (where benefit is robust) with long- $L$  cells (where it degrades); the degradation is  $L$ -conditional, not pooled.

#### 5.5 Sliding window versus exponential decay

Table 5 shows  $L_{\text{max}}$  for both methods. The decay controller does not follow the  $L_{\text{max}} \propto T_{\text{corr}}$  scaling: it achieves  $L_{\text{max}}=67$  at very\_slow (NS,  $L_{\text{max}}$  undefined),  $L_{\text{max}}=5$  at slow, and  $L_{\text{max}}=10$  at mid — an irregular pattern with no consistent constant  $c$ . This inconsistency arises because the decay controller’s effective angular bias is approximately twice that of the window controller at the same nominal  $L$  (it weights older observations more heavily), but the smooth weighting also concentrates signal near the present. The two effects partially offset each other in a  $L$ -dependent and  $\omega$ -dependent way that resists a simple scaling prediction.

**Practical recommendation:** The sliding-window controller provides a *predictable*  $L_{\text{max}} \propto T_{\text{corr}}$  relationship; the exponential-decay controller does not, making it less suitable for principled memory-window selection when  $\omega$  is known or estimated.

**Table 3:  $L_{\text{max}}$  (last  $L$  with  $p < 0.05$ , window) and ratio to  $T_{\text{corr}}$ . The two clean interior points (slow, mid) both give  $L_{\text{max}}/T_{\text{corr}} = 0.30$ . † $\omega=2.5$ : no reliable  $L_{\text{max}}$  — positive benefit persists across the full  $L$  grid without a sharp crossover; see Sec. 5.2.**

| $\omega$ | $T_{\text{corr}}$ | $L_{\text{max}}$ | $\omega L_{\text{max}} dt$ | $L_{\text{max}}/T_{\text{corr}}$ |
|----------|-------------------|------------------|----------------------------|----------------------------------|
| 0.75     | 67                | 67               | 1.00 rad                   | 1.00                             |
| 1.5      | 33                | 10               | <b>0.30 rad</b>            | <b>0.30</b>                      |
| 2.5      | 20                | †                | —                          | —                                |
| 5.0      | 10                | 3                | <b>0.30 rad</b>            | <b>0.30</b>                      |
| 17.0     | 3                 | —                | —                          | —                                |

**Table 4: Pooled  $\bar{\Delta}$  across all  $L$  levels, window method.  $p$ -values are one-sided;  $n$  is total seed- $L$  pairs.**

| $\omega$ ( $T_{\text{corr}}$ ) | $n$ | $\bar{\Delta}$ | SE    | $p$     |
|--------------------------------|-----|----------------|-------|---------|
| 0 ( $\infty$ )                 | 512 | +0.742         | 0.117 | < 0.001 |
| 0.75 (67)                      | 256 | +0.453         | 0.175 | 0.005   |
| 1.5 (33)                       | 256 | +0.500         | 0.191 | 0.005   |
| 5.0 (10)                       | 256 | +0.344         | 0.155 | 0.013   |
| 17.0 (3)                       | 256 | +0.215         | 0.167 | 0.099   |

**Table 5:  $L_{\text{max}}$  for both memory methods. The window method follows  $L_{\text{max}} \approx 0.30 \cdot T_{\text{corr}}$ ; the decay method does not. “—” = no  $L$  achieves  $p < 0.05$ .**

| $\omega$ ( $T_{\text{corr}}$ ) | $L_{\text{max}}$ (window) | $L_{\text{max}}/T_{\text{corr}}$ | $L_{\text{max}}$ (decay) | $L_{\text{max}}/T_{\text{corr}}$ |
|--------------------------------|---------------------------|----------------------------------|--------------------------|----------------------------------|
| 0.75 (67)                      | 67                        | 1.00                             | —                        | —                                |
| 1.5 (33)                       | 10                        | 0.30                             | 5                        | 0.15                             |
| 5.0 (10)                       | 3                         | 0.30                             | 10                       | 1.00                             |
| 17.0 (3)                       | —                         | —                                | —                        | —                                |

#### 5.6 Theoretical prediction vs. empirical result

The pre-experiment bias/noise-parity prediction was  $c \approx 1.0$  (Section 3.4); the empirical value is  $c \approx 0.30$ , a factor of 3 lower. Table 1 shows why: at  $L=L_{\text{max}}$  the ratio  $\beta/\sigma_{\theta,s}$  is 0.38 and 0.21 for slow and mid respectively — far below the  $\approx 1.0$  threshold at which directional MSE parity is achieved. The policy nonlinearity (capacity-matching sends all collectors toward a single estimated direction) means the coordination benefit collapses well before the shared arm’s individual directional accuracy drops to parity with the independent arm. The crossover is driven by coverage loss, not by information loss.

## 6 Discussion

*Practical memory-window heuristic.* For the specific parameter regime studied here ( $\alpha=0.06$ ,  $\sigma=0.06$ ,  $M=4$ ,  $dt=0.02$ , sliding-window controller), setting  $L \lesssim 0.30/(\omega \cdot dt)$  avoids the stale-coordination failure demonstrated in this work. The constant  $c \approx 0.30$  is a two-anchor preliminary value; re-validation is needed before applying it

to different agent counts, field strengths, or coordination protocols. If  $\omega$  is unknown, estimating it from the rate of change of the team velocity estimate and adjusting  $L$  online is a natural but unvalidated extension.

*Connection to SW-UCB.* SW-UCB sets  $L \propto 1/\Delta$  where  $\Delta$  is a breakpoint rate [5]. Our results are consistent with a cooperative analogue  $L \propto 1/(\omega \cdot dt)$  with  $c \approx 0.30$  in the studied setting. The constant is smaller than the single-agent SW-UCB optimum because multi-agent sharing amplifies the consequence of directional bias; whether this analogy holds under other coordination protocols or reward structures is an open question.

*The exponential-decay controller.* Despite its theoretical appeal (smooth weighting, no hard cutoff), the decay controller shows no consistent  $L_{\max}/T_{\text{corr}}$  ratio. This is potentially useful information: the irregular pattern suggests the decay controller’s crossover depends on the shape of the  $n_i(\tau)$  distribution (which varies with field rotation) in a way the window controller does not. A detailed analysis of this interaction is left for future work.

*Anchor robustness.* The slow anchor ( $\bar{\Delta}=+0.94$ ,  $p=0.049$ ) is borderline: a single additional seed going the opposite direction would push  $p$  above 0.05 and shift the reported  $L_{\max}$  from 10 to 5 steps. Furthermore, applying Holm’s step-down procedure uniformly across slow’s eight cells—as was done post-hoc for the fast condition (Section 5)—the adjusted threshold at  $L=10$ ’s rank (4th-smallest  $p$  of 8) is  $0.05/5 = 0.01$ ; the borderline  $p=0.049$  fails this threshold, shifting  $L_{\max}(\text{slow})$  from 10 to 5 steps (ratio  $5/33 \approx 0.15$ ). The mid anchor ( $L=3$ ,  $p=0.004$ ) survives Holm correction (threshold  $0.05/8 = 0.00625$ ). Under strict uniform Holm the two anchors give  $c = 0.15$  and  $c = 0.30$  respectively — no longer a consistent constant. The pre-registered criterion ( $p < 0.05$  per cell, uncorrected) is the basis for the  $c \approx 0.30$  claim; this sensitivity should be viewed as motivation for validation at additional rotation speeds with non-borderline  $L_{\max}$  values rather than adding seeds post-hoc.

*Generality and functional dependence of  $c$ .* The  $c \approx 0.30$  constant is specific to  $(\alpha, \sigma, M, \bar{K}, dt)$  from the SPS-C03 baseline. A back-of-envelope argument for the dominant dependences: the angular noise of the shared arm scales as  $\sigma_{\theta, \text{shared}} \propto \sigma/(\alpha\sqrt{M\bar{K}L}dt)$ , so larger  $\alpha/\sigma$  or larger  $M$  reduces noise and allows the shared arm to remain beneficial to a longer lag before coverage loss dominates. This implies  $c$  is a decreasing function of  $\alpha/\sigma$  and (weakly) of  $M$  — larger or stronger teams tolerate more staleness. The scaling  $L_{\max} \propto T_{\text{corr}}$  should hold for any memory-based cooperative controller with a fixed-direction prior; quantifying  $c(\alpha/\sigma, M)$  requires a parameter sweep outside the scope of this paper.

*Limitations.* (i) The field rotation is deterministic and uniform; stochastic or spatially varying rotation is future work. (ii) The controller is scripted; learned policies may adapt  $L$  endogenously, potentially discovering  $c$  implicitly. (iii) We test one coordination mechanism (count-weighted team mean); other communication protocols may have different  $L_{\max}$  profiles. (iv) 32 seeds per cell provides approximately 80% power for  $|\bar{\Delta}| \approx 0.9$ ; smaller effects at the boundary may require more seeds. (v) The constant  $c \approx 0.30$  is measured at a single parameter point ( $\alpha=0.06$ ,  $\sigma=0.06$ ,  $M=4$ ,  $\bar{K} \approx 8$ ,

$dt=0.02$ ). Its functional dependence on these parameters is not established here; back-of-envelope arguments for the scaling are given in Section 6 but require a systematic parameter sweep to validate. (vi) The two-anchor result ( $T_{\text{corr}} \in \{33, 10\}$ ) is a preliminary pattern, not a law: the borderline slow anchor ( $p=0.049$ ) does not survive uniform Holm correction, and the supplementary  $\omega=2.5$  condition does not conform to the  $c=0.30$  prediction (Section ??). Additional, non-borderline anchors are needed before the scaling constant can be treated as reliable (see Anchor robustness paragraph above).

## 7 Conclusion

We demonstrated a stale-coordination failure mechanism: once the sliding-window memory spans more than  $\approx 0.30$  rad of total field rotation, the shared arm drives all collectors confidently toward a systematically stale direction, eliminating the coverage diversity that makes coordination beneficial. The largest memory window providing a positive benefit scales as  $L_{\max} \approx 0.30 \cdot T_{\text{corr}} = 0.30/(\omega \cdot dt)$  at two interior rotation speeds ( $T_{\text{corr}} \in \{33, 10\}$ ), with a boundary consistent but censored very\_slow point. The theoretical prediction ( $c \approx 1.0$ ) overshoot by a factor of three because the relevant threshold is not directional MSE parity but the point at which shared-arm confidence in a stale direction outweighs the coverage diversity of independent agents — a policy nonlinearity inaccessible to information-geometric arguments.

Within the specific parameter regime studied ( $\alpha=0.06$ ,  $\sigma=0.06$ ,  $M=4$ ,  $dt=0.02$ , sliding-window controller), the heuristic  $L \lesssim 0.30/(\omega \cdot dt)$  avoids this failure mode. The constant  $c \approx 0.30$  is a preliminary two-anchor result; the borderline slow anchor ( $p=0.049$ ) and the anomalous  $\omega=2.5$  condition both motivate further validation before this constant is used prescriptively. The exponential-decay controller does not follow the same scaling, making the sliding window the more predictable choice for memory-window selection under known  $\omega$  in this setting.

## A Full Per-Cell Results

Table 6 reports  $\bar{\Delta}$ , SE, and  $p$ -values for all window-method cells. Decay-method  $L_{\max}$  results are summarised in Table 5 in the main text. Significance at  $p < 0.05$  is marked with (\*).

## B Reproduction Gate Details

Gate run:  $\omega=0$ ,  $L=1$  (stateless sliding window), seeds 6001–6032.  $\bar{\Delta}=+1.188$  (SE = 0.213, 95% CI [+0.77, +1.61],  $p < 0.001$ ), sign count 20/32. This matches SPS-C03’s  $\bar{\Delta}=+1.19$  (SD 2.44, 20/32 positive) within rounding error, confirming the rotating-field implementation is correct at  $\omega=0$ . Note: the reproduction gate uses  $L=1$ , not  $L=H=67$ . The  $L=67$  stationary cell in Table 2 is a separate full-history sliding-window cell that coincidentally yields  $\bar{\Delta}=+1.19$  because field stationarity makes window length irrelevant; it is *not* the gate cell.

## C Proof Sketch: SNR under Angular Bias

At step  $t$ , the window estimate  $\hat{\theta}(t, L)$  (Equation 1) has expected direction  $\hat{\theta} = \theta(t) - \omega L dt/2$  (lag of window midpoint) and sinc-attenuated magnitude  $\alpha \text{sinc}(\omega L dt/2)$ . In the crossover region (cells at or below  $L_{\max}$ , where  $\omega L dt \leq 0.30$  rad) the argument is at most 0.15 rad, giving  $\text{sinc}(0.15) \approx 0.996$  ( $< 0.5\%$  amplitude loss), so sinc

**Table 6: Window method: all 40 non-gate cells ( $n=32$  each).**

| $\omega$  | $L$ | $\bar{\Delta}$ | SE   | $p$   |
|-----------|-----|----------------|------|-------|
| very_slow | L1  | +0.53          | 0.44 | 0.115 |
|           | L3  | +0.47          | 0.67 | 0.243 |
|           | L10 | +0.22          | 0.48 | 0.322 |
|           | L30 | +0.41          | 0.40 | 0.156 |
|           | L67 | +0.78*         | 0.46 | 0.044 |
| slow      | L1  | +1.47*         | 0.51 | 0.002 |
|           | L3  | +1.50*         | 0.41 | 0.000 |
|           | L5  | +1.66*         | 0.45 | 0.000 |
|           | L10 | +0.94*         | 0.57 | 0.049 |
|           | L20 | -0.38          | 0.47 | 0.788 |
|           | L30 | -0.28          | 0.52 | 0.705 |
|           | L45 | -0.47          | 0.59 | 0.786 |
| mid       | L67 | -0.44          | 0.63 | 0.758 |
|           | L1  | +0.88*         | 0.39 | 0.012 |
|           | L3  | +0.91*         | 0.34 | 0.004 |
|           | L5  | +0.22          | 0.41 | 0.296 |
|           | L10 | +0.25          | 0.54 | 0.320 |
|           | L20 | +0.56          | 0.46 | 0.113 |
|           | L30 | +0.16          | 0.50 | 0.378 |
| fast      | L45 | -0.25          | 0.41 | 0.727 |
|           | L67 | +0.03          | 0.43 | 0.471 |
|           | L1  | +0.47          | 0.44 | 0.145 |
|           | L3  | +0.47          | 0.44 | 0.145 |
|           | L5  | +0.78          | 0.39 | 0.023 |
|           | L10 | -0.09          | 0.52 | 0.572 |
|           | L20 | +0.09          | 0.46 | 0.420 |
|           | L30 | -0.38          | 0.49 | 0.777 |
|           | L45 | +0.13          | 0.53 | 0.406 |
|           | L67 | +0.25          | 0.50 | 0.310 |

attenuation is negligible *in this region*. Beyond  $L_{\max}$ , the approximation breaks down: at ( $\omega=5, L=67$ ) the argument is  $5 \times 67 \times 0.02/2 = 3.35$  rad, and  $\text{sinc}(3.35) = \sin(3.35)/3.35 \approx -0.06$ , which is not negligible. However, these cells are already in the degraded regime where  $\bar{\Delta} < 0$ ; the mechanism there is angular bias, not sinc attenuation. The dominant factor throughout is the angular lag  $\beta = \omega L dt/2$ .

The SNR of the direction estimate for  $M$  agents is approximately

$$\text{SNR}_{\text{shared}}(L, \omega) = \frac{\alpha \cos(\beta)}{\sigma / \sqrt{M \bar{K} L dt}},$$

and the ratio to the independent arm is  $\sqrt{M}$ , independent of  $L$  and  $\omega$  — showing the simple model cannot predict negative  $\bar{\Delta}$ . Full derivation available on request.

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